- vector equation of the plane in parametric form is $\vec{r} = \vec{a} + s\left(\vec{b} - \vec{a}\right) + t\vec{c}$, where $s, t \in \mathbb{R}$ which implies that $\vec{r} = \left(-\hat{i} + 2\hat{j}\right) + s\left(3\hat{i} - \hat{k}\right) + t\left(\hat{i} + \hat{j} - \hat{k}\right)$, where $s, t \in \mathbb{R}$.
- vector equation of the plane in non-parametric form is $(\vec{r} - \vec{a}) \cdot ((\vec{b} - \vec{a}) \times \vec{c}) = 0$.

Now, $(\vec{b} - \vec{a}) \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 0 & -1 \\ 1 & 1 & -1 \end{vmatrix} = \hat{i} + 2\hat{j} + 3\hat{k}$,

we have $(\vec{r} - (-\hat{i} + 2\hat{j})) \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 0 \Rightarrow \vec{r} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 3$

- If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is the position vector of an arbitrary point on the plane, then from the above equation, we get the Cartesian equation of the plane as $x + 2y + 3z = 3$.

EXERCISE 6.7

1. Find the non-parametric form of vector equation, and Cartesian equation of the plane passing through the point $(2,3,6)$ and parallel to the straight lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-3}{1}$ and $\frac{x+3}{2} = \frac{y-3}{-5} = \frac{z+1}{-3}$
2. Find the non-parametric form of vector equation, and Cartesian equations of the plane passing through the points $(2,2,1)$, $(9,3,6)$ and perpendicular to the plane $2x + 6y + 6z = 9$.
3. Find parametric form of vector equation and Cartesian equations of the plane passing through the points $(2,2,1), (1,-2,3)$ and parallel to the straight line passing through the points $(2,1,-3)$ and $(-1,5,-8)$.
4. Find the non-parametric form of vector equation and cartesian equation of the plane passing through the point $(1,-2,4)$ and perpendicular to the plane $x + 2y - 3z = 11$ and parallel to the line $\frac{x+7}{3} = \frac{y+3}{-1} = \frac{z}{1}$.
5. Find the parametric form of vector equation, and Cartesian equations of the plane containing the line $\vec{r} = (\hat{i} - \hat{j} + 3\hat{k}) + t(2\hat{i} - \hat{j} + 4\hat{k})$ and perpendicular to plane $\vec{r} \cdot (\hat{i} + 2\hat{j} + \hat{k}) = 8$.
6. Find the parametric vector, non-parametric vector and Cartesian form of the equations of the plane passing through the three non-collinear points $(3,6,-2), (-1,-2,6)$, and (6, 4, –2).
7. Find the non-parametric form of vector equation, and Cartesian equations of the plane $\vec{r} = \left(6\hat{i} - \hat{j} + \hat{k}\right) + s\left(-\hat{i} + 2\hat{j} + \hat{k}\right) + t\left(-5\hat{i} - 4\hat{j} - 5\hat{k}\right)$.

6.8.7 Condition for a line to lie in a plane

We observe that a straight line will lie in a plane if every point on the line, lie in the plane and the normal to the plane is perpendicular to the line.

(i) If the line $\vec{r} = \vec{a} + t\vec{b}$ lies in the plane $\vec{r} \cdot \vec{n} = d$, then $\vec{a} \cdot \vec{n} = d$ and $\vec{b} \cdot \vec{n} = 0$.

(ii) If the line $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$ lies in the plane $Ax + By + Cz + D = 0$, then $Ax_1 + By_1 + Cz_1 + D = 0$ and $aA + bB + cC = 0$.

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